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Optimizing eDNA Replication for Standardized Application in Lotic Systems in Aotearoa, New Zealand

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ABSTRACT

National standardization of ecological sampling protocols between different agencies in Aotearoa New Zealand has historically been difficult to attain, creating challenges for combining datasets for national scale analyses. The introduction of new methods for biological monitoring, such as environmental DNA (eDNA), presents an opportunity to standardize aquatic sampling protocols prior to widespread adoption. The objective of this study was to optimize eDNA sample replication for the consistent characterization of freshwater fish and macroinvertebrate communities in flowing waters, and ultimately, to inform the development of robust national monitoring standards. A comparison of field replication and extraction methods (pooling of preservation buffer) was also trialed as part of this high replication ($n=16$) eDNA study to assess any potential benefits in measuring species richness and reducing processing costs alongside replication optimisation. This involved two 'syringe' sampling methods ('standard' and 'boosted', eight each) conducted across 54 riverine sites throughout the country. No significant difference was found for species richness between the standardized (eight replicates) or boosted (16 replicates composited to eight) eDNA methods for fish and macroinvertebrates. Results indicated that six replicates were needed to consistently detect 89.5% of fish species likely to be present using field-based syringe eDNA sampling and preservation. However, an altitudinal species richness effect was observed for fish. For macroinvertebrates, six replicates were required to identify 86% of taxa identified to the NEMS (National Environmental Monitoring Standards) level used for the Macroinvertebrate Community Index (MCI: usually genera) while eight replicates were required to detect 89% of NEMS taxa. For fish and macroinvertebrate biodiversity, this study suggests that six replicates are a reasonable trade-off between effective community characterization and cost in New Zealand lotic systems.

1 | Introduction

Environmental DNA (eDNA) can be detected in soil, air, and water and its use in aquatic science has rapidly increased since first being introduced for species detection in water in the late 2000's (Ficetola et al. 2008). The collection of eDNA, especially suited to rivers and streams as genetic material carried downstream, is regularly mixed and suspended and transported in a

similar manner to fine suspended sediment (Pont et al. 2018). For eDNA, metabarcoding DNA is usually processed by using polymerase chain reaction (PCR) primers and next-generation sequencing (NGS) platforms resulting in rapid and accurate bio-assessments of often complex aquatic communities (Valentini et al. 2016; Wilkinson et al. 2024). Other more targeted and often more sensitive PCR methods (qPCR and ddPCR) are also available depending on the study objectives.

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Detection of eDNA in lotic systems is dependent on many variables (Barnes et al. 2014; Barnes and Turner 2016), including its rate of production and degradation, deposition, resuspension, dilution, transport (Carraro et al. 2018), and season (Furlan et al. 2016). eDNA degradation may increase with UV irradiance (Ravanat, Douki, and Cadet 2001; Pilliod et al. 2014; Strickler, Fremier, and Goldberg 2015), temperature (Zhu 2006; Lacoursière-Roussel et al. 2016; Kasai et al. 2020), and microbial consumption (Salter 2018). These variables will likely alter from one system/site to the next, with changes in land use, elevation, geology, substrate (Jerde et al. 2016), and flow (Shogren et al. 2018). Spatial and temporal longitudinal changes in biological communities are also likely to occur (Yamamoto et al. 2017; Stoeckle, Soboleva, and Charlop-Powers 2017; Bylemans et al. 2018) from estuaries to headwater streams. This is especially true in Aotearoa New Zealand (NZ) where elevation and distance inland have a strong influence on native fish diversity because of high levels of diadromy (Hayes, Leathwick, and Hanchet 1989; McDowall 1993; Jowett and Richardson 1996).

eDNA metabarcoding in many cases has proven to be a powerful tool for describing fish (Taberlet et al. 2012; Civade et al. 2016; Yamanaka and Minamoto 2016; Nakagawa et al. 2018; Pont et al. 2018) and invertebrate (Deiner and Altermatt 2014; Fernández et al. 2019) communities. However, one limitation of the eDNA method includes the generation of false negatives (non-detection when present), which occur more frequently when insufficient sample replication is used (Ficetola et al. 2015; Roussel et al. 2015), or when insufficient water is filtered per replicate (Mächler et al. 2016; Cantera et al. 2019). eDNA in water tends to be patchily distributed and differentially degraded, and therefore, sample replication and volume are important factors for reducing the potential of false negatives. However, higher replication increases study costs, and possible false positives due to more samples being collected and the potential for contamination during handling associated with these extra samples (Ficetola et al. 2015). Thus, optimizing sample replication is one of the most critical steps in the design of a robust eDNA biodiversity study (Yoccoz 2012; Takahashi et al. 2023).

Prior to the development of eDNA analysis' aquatic biological indicators for assessing ecosystem health, such as the index for biotic integrity (IBI) for fish (Karr 1981) or the Macroinvertebrate Community Index (MCI), (Stark and Macted 2007) have historically been developed using more conventional sampling methods (e.g., electric fishing, trapping, netting for fish surveys, and kick sampling for aquatic invertebrates). These indices are used to evaluate various aspects of biological community structure and function, such as for evaluating the presence/absence of native or introduced species, population structure, or assessing environmental effects of a range of stressors.

eDNA offers a relatively new and promising method to monitor and assess freshwater systems and to complement conventional sampling approaches. The use of eDNA has many benefits and advantages compared to conventional lotic sampling techniques, in that little training is needed, and in many cases, eDNA sampling is more cost effective (equipment, staff and time required is minimal) and less invasive (Sigsgaard et al. 2015; Qu and Stewart 2019). There is no need for taxonomists to identify specimens

(Uchida et al. 2020), and there is improved detection of cryptic species (Yao et al. 2022) and a wider range of taxa (Takahara et al. 2012; Civade et al. 2016; Deiner et al. 2017; Pont et al. 2018). Accessing and sampling a site is relatively straight forward with eDNA (Reinhardt et al. 2019) as only one access point is required for well mixed streams compared to at least 150m for conventional methods (Joy, David, and Lake 2013) in NZ which can be an issue for smaller streams with heavy riparian vegetation and/or non-wadable systems. This "single access" sampling may require further thought for larger less mixed lotic systems. Additional ecological information or bycatch (non-target) data, both aquatic and terrestrial, can also be used for conservation and biosecurity purposes. eDNA, however, does have several limitations, such as size structure, fish health, and abundance which cannot be obtained but are readily obtained by traditional methods.

While standardized sampling protocols have been established for conventional fish (Joy, David, and Lake 2013) and macroinvertebrate sampling (NEMS 2022), in NZ and elsewhere, a similar standardized approach is also needed for eDNA sampling. This is particularly true where eDNA data is used for national level reporting, assessments of environmental effects (AEEs), and biosecurity surveillance. A standardized national approach also helps with the development of ecosystem health indices and benchmarks. In this study, standardization mainly refers to the amount of field replication needed rather than the entire eDNA workflow used where these replication levels can be applied irrespective of downstream steps. In this paper, we aim to progress the preliminary work of David et al. (2021) to standardize eDNA sampling in NZ streams and rivers. By critically analyzing high replication data from a broad range of lotic systems, the intent was to optimize sample collection and pooling to enable consistent, comparative, and robust species richness assessment at the national scale.

2 | Materials and Methods

2.1 | eDNA Collection and Analysis

Fourteen of sixteen NZ regional councils and two unitary authorities contributed samples to the study. Each council contributed one to six sites for sampling, with 54 sites being sampled. This nationwide approach enabled a wide range of species and sites to be sampled with differences in land use and geology. Sites ranged, from near sea level (2.2m) to 600m in elevation and from small first order streams to NZ's longest river (The Waikato), a 7th order site and NZ's largest river by volume (The Clutha), an 8th order site. Baseline invertebrate data (kick net) were available for all but one of the sites sampled ($n = 53$), and six sites also had electric fishing data (single pass, 150 m) for comparison with the eDNA results. At each of the 54 sites (Table A1), 24 replicate eDNA water samples were collected using Wilderlab V2 eDNA mini kits (each containing a 1.2- μ m, 30 mm cellulose acetate syringe filter, and 350 μ L Zymo 2X DNA/RNA shield for filter preservation and sample lysis). Fieldworkers entered wadable stream sites from a downstream direction standing parallel to each other (when in pairs or larger teams) and filtered stream water from a flowing section of the stream (normally the middle of a riffle section) rather than a

backwater or slow flowing pool. For larger waterways (stream order > 6), streams were again waded and sampled in flowing water but closer to the margins rather than the middle of the stream/river where depth and velocity made wading difficult. All 24 field replicates were collected from that single location for each site. At each site, new latex gloves were used along with a sterile 60 mL syringe to filter surface water through the encapsulated filter membrane for the collection of biological material 50 mL at a time. After 1 L of water was filtered (or the filter became clogged, whichever came first), 60 mL of air was pushed through the filter using the same syringe to remove excess water from the filter. Each sample (1 L) took between 5–10 min to filter depending on turbidity. Eight sites out of the 54 were filtered till clogged due to higher levels of turbidity. The Shield preservative (Zymo Research Cat. No. R1200-125) was then injected directly into the filter for metazoan cell lysis and to prevent degradation of captured DNA, and the outlet end of the filter sealed using a luer-lock syringe cap (so no liquid can spill out). This was repeated for each replicate. Eight of the replicates were randomly separated and labeled as “Standard” while the remaining 16 were labeled “Boosted”. These 16 “boosted” samples were randomly paired and combined on receipt at the Wilderlab laboratory to make eight (600 µL) “boosted” samples. Each of these eight “boosted” samples is a composite of two “standard” replicates or the pooling of the preservation buffer of two “standard” replicates to make one “boosted” sample. The total volume filtered and GPS co-ordinates of the location were recorded on the supplied sample kit bag. This same process was repeated for all sites, and samples were kept at ambient temperature and couriered to the Wilderlab laboratory (Wellington, NZ) within one week of collection for analysis. No negative field control samples were taken in this study, and in hindsight, some should have been taken. However, from the 51 sites assessed in this study, no fish species was found where it should not be. More recently obtained data (J. Smith, unpublished data) using the standard six replication method with some negative field controls did show some minor contamination (low sequence reads) for both fish and invertebrates for species that occurred at sampling sites. We suggest that this is likely caused by aerosolised biological molecules from the waterway entering the water used for the negative control.

On receipt of the preserved filters at the laboratory, the 8 “standard” kits were processed as normal, where the shield buffer was drawn from the capture filter using the attached 3 cc syringe and transferred into clearly labeled individual LoBind Eppendorf tubes. The volume of shield that can be drawn from the filter varies but is generally in the vicinity of 300 µL; then, 200 µL is transferred to the extraction cartridge. The shield is drawn off the filter, rather than digesting the filter directly, since the Zymo DNA/RNA shield is a lysis buffer (https://files.zymoresearch.com/protocols/_r1100-50_r1100-250_r1200-25_r1100-125_dna_rna_shield.pdf), so target metazoan DNA is in solution by the time the lysate is extracted from the filter.

The ‘boosted’ samples were treated in the same way after pooling to create 8 × 600 µL samples. All samples were stored in a –20°C freezer until further processing. The indirect extraction method compared favorably to direct filter digestion in a previous study (Banks, Kelly, and Clapcott 2020) noting that the extractions

were carried out in different laboratories. Filter capsules are not cracked to minimize the risk of contamination.

2.1.1 | DNA Quality Analysis and NGS Library Preparation

For DNA extraction and purification, 200 µL of lysate from each sample was loaded into a Genolution GD141 DNA/RNA extraction kit cartridge and run on the Genolution Nexttractor NX-48S system using the standard (default) extraction settings (see Colman et al. 2020, for more on Genolution Nexttractor NX-48S). The same volume of eluate for both “standard” and “boosted” samples was used in the amplification process (200 µL). DNA quality/quantity analysis, adapter-fusion, indexing, and amplification were carried out in single-step quantitative PCR (qPCR) reactions on an Applied Biosystems QuantStudio 1 qPCR instrument. DNA extracts were PCR-amplified (singleplex for each primer pair) using the Wilderlab comprehensive freshwater multi-species analysis package (see Table 1 for list of primers). In total, six assays were used to detect fish and eight primers were used to detect NEMS (MCI) taxa (Table 2). Fusion-tagged forward primers were designed as follows: P5 Illumina adapter sequence + Illumina TruSeq sequencing primer + 8-bp unique index tag + locus specific forward primer from Table 1. Fusion-tagged reverse primers were designed as follows: P7 Illumina adapter sequence + 8-bp unique index tag + locus specific reverse primer from Table 1. All 8-bp index tags differed from each other by at least 3 bp. Each reaction contained 5 µL of SensiFAST 2× LoRox SYBR Mix (Bioline), 0.25 µL forward primer (10 µM), 0.25 µL reverse primer (10 µM), 0.5 µL BSA (10 mg mL⁻¹, Sigma Aldrich), 2 µL de-ionized water (nuclease-free), and 2 µL template DNA. qPCR cycling conditions included an initial denaturation of 3 min at 95°C; followed by 40 cycles of 5 s at 95°C, 10 s at 58°C (WV and RV assays), 45°C (CI assay) or 52°C (all other assays), and 15 s at 72°C. There was no multiplexing, and all PCRs were run separately with a single pair of primers per reaction. Aside from the annealing temperatures (see Wilkinson et al. 2024), cycling conditions were the same for each assay. DNA quality and quantity were assessed by checking that a sigmoidal log-amplification curve was visible with a CT value < 35. A negative control reaction containing 2 µL of de-ionized water in place of the template DNA was included with each qPCR run. The blank controls are included as raw fastq files in the SRA archive PRJNA950216 (<https://www.ncbi.nlm.nih.gov/bioproject/950216>). None of the blanks contained identifiable fish sequences at levels above the sequence count threshold of 5. This threshold is set to account for PCR misreads and sequencing errors; the forward and reverse indexes are all 8 bp and all differ from each other by at least 3 bp, but there still exists a low sequencing misread/PCR error rate that can occasionally lead to sequences being assigned to the wrong sample during demultiplexing (tag jumping; see Schnell, Bohmann, and Gilbert 2015). We find that a minimum sequence count of 5 prevents false positives from PCR misreads and sequencing errors at the sequencing depth at which we operate. The low-pass sequencing method we present here was designed to minimize the cost benefit ratio for the purposes of fish monitored by Regional Councils in NZ, and we have shown in a previous paper (David et al. 2021) that this approach can outperform traditional monitoring methods for this purpose (i.e.,

TABLE 1 | Assays included in the Wilderlab comprehensive freshwater multi-species package, with sequences for forward and reverse primers and average fragment length (note, WV is generally 78 bp for fish).

Assay	Gene	Forward primer	Reverse primer	Av fragment length	Reference
BE	18S-V9	CCCTGCCHTTTGTACACAC	CCTTCYGCAGGTTTCACCTAC	100–150	[1]
BU	18S-V9	TTGTACACACCGCCC	CCTTCYGCAGGTTTCACCTAC	80–150	[1]
CI	COI	DACWGGWTGAACWGTWTAYCCHCC	GTTGTAATAAAATTAAAYDGCYCCTARAATDGA	76	[2]
DG	mt16S	TCTTCGGTTGGGGCGAC	GGATTGCGCTGTTATCCCT	40–150	[3]
LV	12S	TCGTGCCAGCCRCGC	CATAGTGGGTATCTAATCCCAGTTTG	100–165	[4]
MZ	rbcL	CTTCTTCAGGTGGAACCTCCAG	GTCACCAACAACAGAGACTAAAGCAAGT	105	[5]
RV	12S-V5	TTAGATACCCCACTATGC	TAGAACAGGCTCCTCTAG	97–103	[6]
TP	trnL	GGCAAATCCTGAGCCAA	CCATTGAGTCTCTGCACCTATC	35–150	[7]
UM	16S-V5	GGATTAGATACCCTGGTA	CCGTCAATTTCMTTTRAGTTT	100–105	[8]
WV	mt16S	GACGAGAAAGACCCTWTGGAGC	CCRYGGTCGCCCCCAAC	35–150	[9]
ZC	mt16S	GGACGATAAGACCCCTATAAADCTT	CGCTGTATCCCTAAAGTAAAYTT	80–150	[10]

Note: [1] Forward and reverse primers from (Amaral-Zettler et al. 2009); [2] forward primer adapted from (Vamos, Elbrecht, and Leese 2017); reverse primer developed in-house at Wilderlab; [3] forward primer developed in-house at Wilderlab; reverse primer adapted from (Miya et al. 2015); [5] forward and reverse primers from (Bradley et al. 2007); [6] forward primer adapted from (Riaz et al. 2011); reverse primer from (Kelly et al. 2014); [7] forward and reverse primers from (Taberlet et al. 2007); [8] forward primer from (Morey et al. 2006); reverse primer from (Lane et al. 1985); [9] forward and reverse primers adapted from (Nester et al. 2020); [10] forward and reverse primers developed in-house at Wilderlab.

electrofishing). Aside from increasing the costs, which leads to lower uptake of the tool, increasing the sequencing depth also increases the number of PCR misreads, which leads to higher minimum count thresholds and other stringent controls such as minimum occupancy (i.e., a species must appear in more than one replicate to be considered a ‘true positive’). The assays in this panel range from specific to generalist, and for more specific assays, it is very difficult to get the number of sequence reads above 5000 when the target organisms are at low prevalence in the sampling area. For example, the RV ‘ecoprimer’ assay is very specific for vertebrates and is virtually impossible to reliably get > 5000 reads at reference sites where the incidence of fish is low. Next generation sequencing (NGS) libraries were pooled at roughly equimolar concentration using the final normalized ΔR_n fluorescence values as a guide and cleaned and double-end size selected using AMPure XP magnetic beads (0.9 $\times$ and 1.2 $\times$ for lower and upper size bounds, respectively). The final pooled library concentration was determined using a Qubit 4 Fluorometer (ThermoFisher Scientific), and the concentration was adjusted to 50 pM in sterile DNase/RNase free water. The library was then loaded onto an iSeq i1 V2 reagent cartridge with 300-cycle flow cell (Illumina) with 5% Phi X and run for 290 cycles in a single direction on an Illumina iSeq 100 instrument.

TABLE 2 | eDNA primers used for fish and NEMS taxa.

MCI taxa	Fish
WV	RV
DG	LV
CI	WV
BE	DG
BU	CI
ZC	ZV
ZV	
LV	

2.1.2 | Sequence Depth Analysis

A key variable requiring optimization in metabarcoding analysis involves sequencing depth, or the number of reads per assay per sample. Too little depth/coverage can increase the false negative rate, as DNA signals from rare species may not appear in the ASV profile or fall below the minimum sequence count threshold. Excessive sequencing depth can inflate costs, requiring larger sequencing platforms with significantly more expensive reagent kits. This inevitably comes at the cost of reduced field replicates, sites to include, or the number of metabarcoding assays to run. As such, it is important to find the point of diminishing return as the number of reads at which the number of species detected in a sample levels off and temper sequencing depth accordingly.

Using the core fish assays (mitochondrial 12S ‘RV’ and 16S ‘WV’) and invertebrate assay (COI ‘CI’), we ran species accumulation profiles within each sample where over 5000 sequence reads were available, by subsampling the sequences to 100, 200, 300, ... 4900, and 5000 reads. Species accumulation plots with 95% confidence intervals were generated using the ggplot2 package in R (Wickham 2016; R Core Team 2021), and the sequencing depth (no. of reads per sample per assay) at which 95% of the full target taxon profile was detected was interpolated from the plots. For the WV and RV assays, the target taxon profile was restricted to fish resolved to species level, and for the CI assay, the target taxon profile was restricted to insects resolved to the genus or species level (since most MCI indicators require identification to the genus level).

All species accumulation curves had leveled off at 5000 sequence reads, with the vertebrate-specific 12S ‘RV’ assay requiring only 500 reads on average to capture 95% of the fish species present in the sample. As expected, the more generalist CI assay had a flatter curve, requiring 2500 reads on average to account for 95% of the insect genera and species in each sample. The WV assay, which was designed for vertebrates but also detects several worms and other inverts, had an intermediate species accumulation curve, requiring 1000 reads to reach 95% fish species saturation (Figure 1).

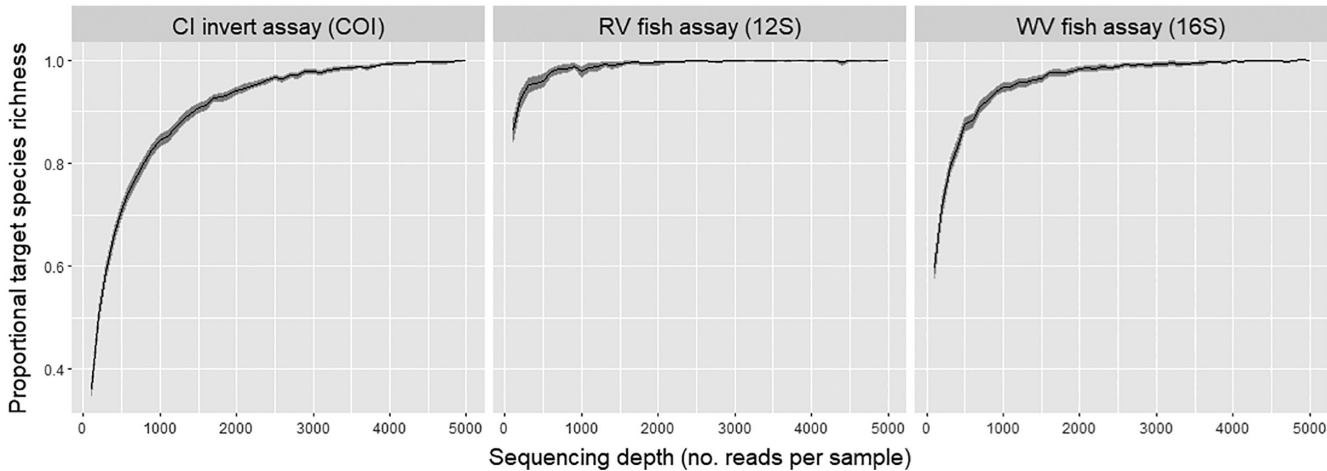


FIGURE 1 | Species accumulation curves by sequencing depth for the core fish (RV and WV) and insect (CI) metabarcoding assays used in this study.

Our average read depths for the three assays covered here were well above their 95% saturation limits, with 4444, 11,377, and 7621 reads per sample for the RV, WV, and CI assays, respectively (see Wilkinson et al. 2024; Table 1).

These results are consistent with those of García-Machado et al. (2023), who found that filtering ASVs to only retain those representing at least 1/1000 of the target taxon profile was optimal to remove false positives and retain true positives in the data.

2.1.3 | V Generation and Taxonomic Assignment

The iSeq 100 output sequence FASTQ files were de-multiplexed in R (R Core Team 2021) using the insect package (v 1.4.0; Wilkinson et al. 2018), and trimmed sequences were filtered to produce a table of exact amplicon sequence variants (ASVs) using the DADA2 R package (Callahan et al. 2016). Sequence filtering was done using the DADA2 R package to remove PCR errors and cameras using a standard model-based approach. The ASVs produced by this approach are considered to be the true sequences present in the sample (though a 100% error removal rate is not strictly possible). ASVs were identified to the lowest possible taxonomic rank using Wilderlab's four-step classification process. This involves: (1) exact matching against an intensively-curated database of previously detected eDNA sequences; (2) remaining (i.e., non-matched) sequences are exact-matched against a global reference sequence database primarily compiled of trimmed reference sequences downloaded from GenBank (Benson et al. 2010) and BOLD (Ratnasingham and Hebert 2007) and then matching sequences assigned at the lowest common ancestor level (LCA; i.e. assigned to the genus level if matched with 100% identity to more than one species or to family level if matched to more than one genus); (3) the remaining sequences that are > 50bp in length are matched with single indel/substitution tolerance against the same GenBank/BOLD reference database and matching sequences assigned at the LCA level; and (4) the remaining sequences are queried against the local GenBank/BOLD reference database using the SINTAX classification algorithm (Edgar 2016) with a conservative assignment threshold of > 0.99 (all rank levels) and maximum genus level assignment threshold. Verification of previously detected eDNA sequences involves initial classification using publicly available reference data (i.e., GenBank) using matching and SINTAX classification (as above) and then checking the distribution of each unique eDNA sequence against the known distribution of the taxon it has been assigned to. If an eDNA sequence that has been assigned to a species repeatedly appears outside the known range of the species, this suggests the presence of a congeneric with an identical barcode sequence that is missing from the reference database and causing an over-classification error (Edgar 2018). In this case, the taxon identification of the eDNA sequence can be manually assigned to a higher rank level, which it will continue to be called at if encountered in future eDNA surveys. This manual curation process is a means to include geographical information into the taxon assignment process and accounts for missing reference sequences in GenBank, which can lead to over-classification and mis-classification errors. This manual curation step is carried out exhaustively for fish, since fish distribution records are available through fishbase and other repositories, and a limited

number of freshwater fish species are present in New Zealand. We do this to a lesser extent for other taxa based on the availability of comprehensive distribution records. The in house reference database used for initial assignment of new eDNA sequences is predominantly composed of GenBank data that is retrieved, trimmed (using a virtualPCR approach with the same primers sequences used in the analysis), and clustered at 97% similarity in order to spot any spurious reference sequences that is labeled as a completely different taxon to the other members of the cluster (using the purge function of the insect R package with default parameters).

2.2 | Data Analysis

Three of the 54 sites were excluded from the data analysis due to field sampling inconsistencies (sites 29 and 44) and heavy rain prior to sampling reducing DNA on the filters (site 13, Table A1). Sequences identified to the species level for freshwater fish and macroinvertebrates were used to generate separate species accumulation plots with quantiles using R, version 4.0.4 (R Core Team 2021). R packages used for data analysis include readxl (Wickham and Bryan 2022), tidyr (Wickham and Girlich 2022), dplyr (Wickham et al. 2022), and BiodiversityR (Kindt and Coe 2005). Accumulation plots were produced using ggplot2 (Wickham 2016) and compared for both the standard and boosted methods. To maintain randomness, the order the samples were collected in the field (random) was the same order the samples were analyzed for the accumulation plots. The proportion of species detected at each site from each replicate is a proportion of the total number (100%) of species detected from all 16 replicates taken (8 single and 8 boosted). This total is the total number of species/taxa detected from eDNA sampling only. Recreational fishers are known to discard marine fish frames (skeletons) into streams (often to feed the local eels); therefore, any truly marine fish species detected by eDNA were excluded from the data analysis. Though rare in this study, any marine “wandering” species that inhabit both marine and freshwater systems were left in the analysis. Although eDNA allowed freshwater invertebrates to be identified to the lowest taxonomic level available in the GenBank/BOLD reference database (usually to species), this data was reassigned to the same taxonomic level as the National Environmental Monitoring Standards (NEMS) invertebrate taxa list (NEMS 2022), which recognized 197 MCI taxa. This NEMS list included 138 genera, 42 families, sub-families or tribes, and 17 higher taxonomic levels (order, class, or phyla). We refer to this simplified list as the “NEMS taxa list”, which is the level of resolution used by regulatory monitoring agencies throughout NZ. Reducing the level of resolution of the eDNA to that used in NEMS meant that any assessment of taxonomic resolution plots made in this study should mimic the same level of taxonomic resolution used throughout the country. As such, we acknowledge that we are probably underestimating the real effect of sample replication on species richness, but that this analysis has more relevance to regional council ecologists.

Single-factor ANOVA was used to compare species richness between the two sampling methods (standard and boosted). Sites that were “filtered to clogged” were also checked separately as a group (8 sites) to check if less water filtered at more turbid sites may have resulted in any differences between the methods used.

Dominant land cover of each site 5 km upstream of the sampling site was established using the New Zealand Land Cover database (LCDB v5., Mainland, New Zealand <https://iris.scinfo.org.nz/>). Elevation, NZ segment, NZ reach, and stream order for each site was derived from the NIWA (National Institute of Water and Atmosphere) REC2v5 (River Environment Classification NZ) dataset [River Environment Classification \(REC2\) NZ](#).

Analysis of stream order for fish and NEMS taxa was achieved by sorting sites into two stream order groups of small sites ($n = 23$), with a stream order of 3 and below, and large sites ($n = 28$), with a stream order of 4 and above [Land, Air, Water Aotearoa \(LAWA\)](#). Similarly, for elevation, two groups were sorted based on sites with an elevation of less than 100 m ($n = 38$) and sites at elevations greater than 100 m ($n = 13$, Table 3). ANOVA was also used to see if invertebrate taxonomic richness differed between sampling sites when assessed using eDNA (using both the original level of identification from the eDNA results, and the eDNA data amalgamated to the NEMS level of identification), and traditional kick net samples, which also identified to the NEMS level. Although the kick net samples were not collected concurrently with eDNA sampling, the median time between sampling with both methods was just over 3 months. Such a short time difference between samples was considered relatively insignificant, especially given the overall persistence of invertebrate communities in New Zealand, and the lack of strong seasonal patterns (Scarsbrook 2002).

3 | Results and Discussion

3.1 | Summary of eDNA Detected

From 51 national sites (Figure 2), 31 different taxonomic groups are represented in the sequence reads ranging from the largest “other” to the smallest “lizards” (Figure 3 and Table A2). The “other” grouping comprises 33.5% of the total reads of all sites and mainly consists of unidentified sequences which are a combination of organisms not currently in the reference database or are undescribed. Freshwater fish represented 11.4% of the total reads. The NEMS taxa include a variety of groups (crustaceans, worms, flatworms, insects, springtails, and mollusks) making up the NEMS taxa list (NEMS 2022) and equates to 13.9% of the total reads. 33 terrestrial mammals (excluding humans and zoo species) and 48 bird species were detected in this study. Several of these species, as are species in other groups (lizards, frogs, plants, and freshwater mussels) are of interest for either conservation or biosecurity. For more information on “other species”

including replication analysis for mammals and birds, please see Appendix S1.

3.2 | Effects of Replication

Accumulation plots were prepared for the two target groups (fish and NEMS taxa) comparing the two methods (standard and boosted, Figure 4). To reach 89.5% and 86% richness (standard method) for fish and NEMS taxa respectively, six replicates are needed. Further two replicates were required to reach 89% richness for invertebrates (Figure 4). As there was no significant difference between the two methods (standard and boosted), all 16 replicates can effectively be combined to assess typical levels of confidence (90% and 95%); for fish, this is 7 and 11 replicates, respectively, and 9 and 11 replicates, respectively, for NEMS taxa. Variation at the 95th percentile for both groups is reduced with increased replication (Figure 4). Fish, however, have the largest reduction in variation initiating at 69.2% for one replicate, which reduces to 27.7% at six replicates, a 41.5% improvement (standard method). In comparison, variation for NEMS taxa reduces by 11% over six replicates indicating similar species richness between sites. As two different sampling methods were used in this study, the asymptote at eight replicates never reached 100% because of the different species detected between methods (standard and boosted). The species proportional richness range for both fish and NEMS taxa improved with each replicate sample. The accumulation plots for both were similar, other than the different starting points or first replicate, for fish at one replicate (standard method) species richness was at a mean of 64% compared to NEMS taxa (57%). After six replicates, the additional increase (relative to one replicate) for fish and NEMS taxa was 25.5% and 29%, respectively, resulting in 89.5 and 86.0 mean percent species richness for these two groups. For fish, the greatest gains in mean species richness occurred within the first four replicates before species richness started to asymptote at around 6 replicates. Invertebrates mirrored fish but at a slightly lower starting point (57% compared with 64%) which is maintained over the first five replicates and narrows to 3.5% after 6 replicates (86% compared to 89.5%). The replication difference for fish and NEMS taxa (species occurrence) over all 16 replicates is summarized at the end of Table 3.

3.3 | Standard Versus Boosted Method

There was no significant difference for species richness between the two methods (standard and boosted) for both fish and NEMS

TABLE 3 | Mean species/taxa proportional richness (%) for fish and NEMS taxa at one and six replicates (standard method) and by stream order and elevation. The last two columns are the number of species per replicate (16 replicates) with range (minimum to maximum) and average.

Group	Standard		Stream order				Elevation				Number of species/rep	
	$n = 51$		1–3, $n = 23$		4–8, $n = 28$		> 100 m, $n = 13$		< 100 m, $n = 38$		$n = 16$	
	1 rep	6 reps	1 rep	6 reps	1 rep	6 reps	1 rep	6 reps	1 rep	6 reps	min–max	Av
Fish	64.0	89.5	65.3	90.6	64.1	89.5	70.4	91.6	61.9	88.8	0–8	3.2
NEMS taxa	57.0	86.0	54.8	87.1	59.5	85.6	57.0	88.6	57.5	85.1	3–17	9.9

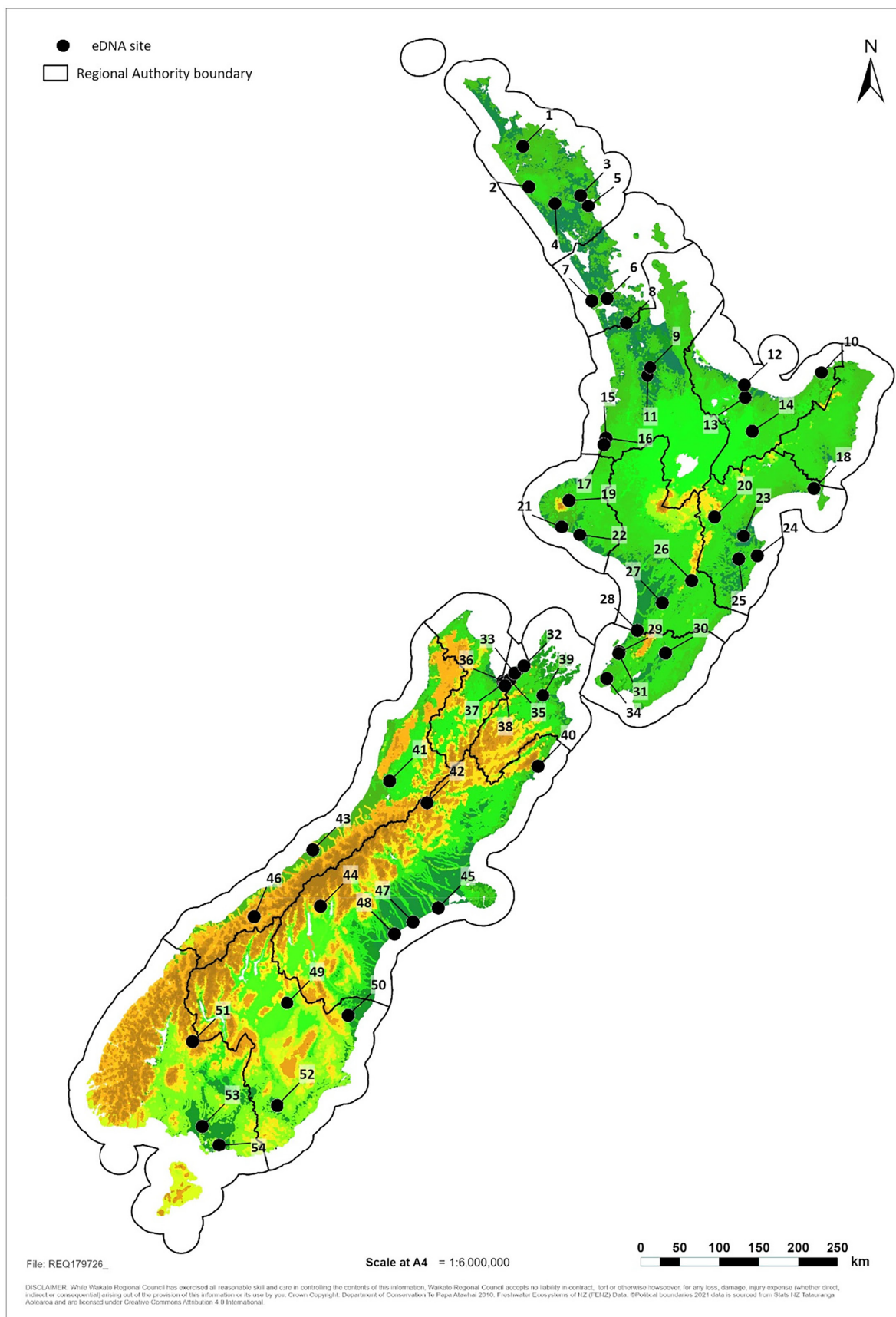


FIGURE 2 | National high-replicate eDNA sites ($n=54$) where samples were collected from December 2020 to May 2021.

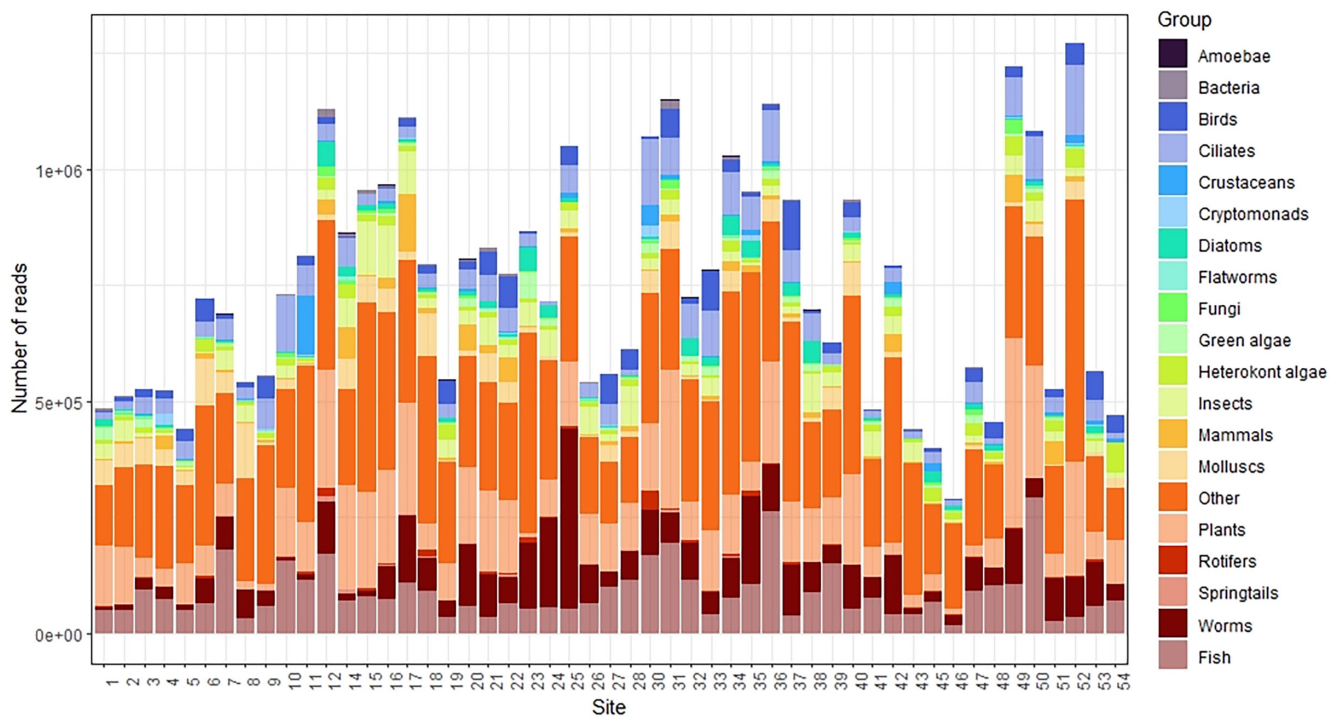


FIGURE 3 | Total eDNA reads by group for all sites analyzed ($n = 51$).

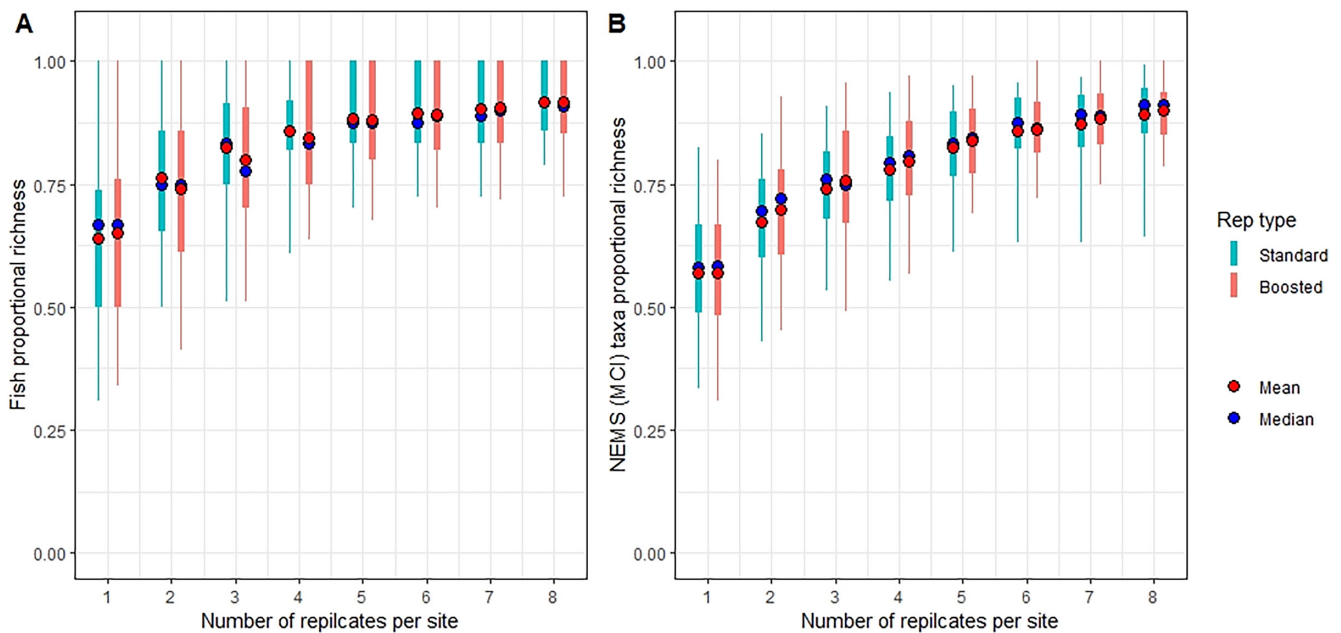


FIGURE 4 | Proportion of taxa richness detected by eDNA sampling for (A) freshwater fish and (B) NEMS (MCI) taxa with standard versus boosted method. Box and whiskers are 50th and 95th percentiles. Red dot = mean and blue dot = median. Proportion of taxa richness for 16 replicates = 1.0.

taxa at 6 or 8 replicates (ANOVA, $df=1$, $F=0.08$, $n=3264$, $p=0.769$). When this study was conceived, the ‘boosted’ method (pooling two samples prior to DNA extraction) was considered as a possibility to improve species detections with less laboratory processing costs. Increasing filtration volumes can improve eDNA yields and species detection rates by promoting the capture of low-concentration biological material from rare species (Hunter et al. 2019). However, filters tend to clog quickly in turbid environments, and the only options to increase filtration

volumes are to either switch to a larger pore size (and risk losing more material through the pores) or increase the area of the filter membrane, which then requires more preservative to inactivate DNA breakdown. In the latter case, the concentration of DNA in the preservative reaches a limit since it is countered by the dilution effects of adding more preservative. Given that only a limited amount of the preservative can be carried through to the DNA extraction step, increasing the effective filter membrane area to increase filtration volume (including pooling samples

prior to extraction) is subject to diminishing returns. Both the standard and boosted samples were collected at the same time and taken from the same parcel of water, but the extra water filtered via the boosted samples provided no advantage in detecting more species. There are differences in results with species detected and sequence read counts between replicates and between the two methods at most sites, reflecting the stochastic nature of eDNA sampling (Cannon et al. 2016; Kawakami et al. 2023), but as a whole, no benefit from using the boosted method was detected.

3.4 | Turbid Sites

At eight of the 51 sites (15.7%), the targeted amount of water filtered (one liter for standard and two liters for boosted) could not be achieved due to excessive turbidity. At these sites, a relative doubling of water filtered from standard to boosted was still achieved (Table A1). These sites were run separately as a group to check for any significant difference in species richness between the standard and boosted methods at 6 replicates and again none was found (ANOVA $p=0.650$ for fish, $p=0.786$ for NEMS invertebrates, df for all = 1, n for all = 128). Filtering such large volumes of water in NZ waterways can require the use of very coarse filters, which in our experience have not yielded improved detection rates over the smaller volume and fine-membrane methods used here (Bird et al. 2024). This is likely due to the loss of target molecules through the pores of the coarse filters, and the accumulation of other debris such as sediment and phytoplankton (Stoeckle et al. 2022). Due to the varied water clarity (high sediment and/or algal loads) of many of NZ's streams, filtering large volumes of water is regarded as somewhat impractical (Bruno et al. 2017), whereas the eDNA mini kits used in this study only require small volumes of water to be filtered (up to 1 L). For sites with higher levels of turbidity, the use of a pre-filter or large filter size could be considered.

3.5 | Fish

For the 51 sites analyzed, 40 of 78 (51.3%) recognized freshwater fish species were detected using eDNA in this study (31 native species and 9 introduced). Eight fish had a conservation status of "threatened" and 10 were "at risk" (Dunn et al. 2018). Fish species not detected were either rare, or geographically isolated to specific areas not covered in this study. For instance, within the family *Galaxiidae*, 13 of the 35 galaxiid species were taken into account in this study, mostly widely distributed species. Most of the undetected galaxiids are naturally uncommon or relict populations residing in small specific geographical areas, mainly in NZ's South Island. The situation was similar for 14 introduced species (4 *Gobiidae*, 4 *Poeciliidae*, 4 *Salmonidae*, and 2 *Cyprinidae*), with all being either rare to occasional, isolated, or both (McDowall 2000). In contrast, all the widely distributed (nationwide) freshwater fish species of NZ were represented in the eDNA results of this study. These distributions varied between species and ranged from common to rare or patchy. For example, several rarer species were only detected at one site (Table A3), whereas longfin eels (*Anguilla dieffenbachii*) were present at all sites. Longfin eel occurrence detection was 94% (standard method) after the

first replicate at the sites it occurred (all but one) indicating its generality, its large size, and status as the apex predator (likely elevating its chance of detection) in NZ freshwater systems and reflecting its ability to penetrate far inland despite being a diadromous species (McDowall 1978). Shortfin eels (*Anguilla australis*) were the second most common fish species in occurrence and were found at 43 sites. Their presence was patchy at some sites, occurring 81% of the time after two replicates and 88% after 5 replicates using standard methods. Shortfin eels are generally more common than longfin eels but tend to prefer warmer lowland waters (Jellyman 1974) and finer substrates at these locations. The Waikato River site at Hamilton (site 11, Table A1, Figure 3) had the highest fish diversity with 17 species, eight of which are introduced species (the highest of any site). While this site is approximately 115 km inland, it remains at low elevation (< 20 m) and is easily accessible by diadromous species. Additionally, the warmer northern waters of the lower Waikato River are a hotspot for invasive fish (Collier and Grainger 2015). Fish species richness ranged from 2 to 17 across all sites. On average, across all 16 replicates, fish richness was 3.2 species per replicate and ranged from 0 to 8 per replicate (Table 3).

Single pass electric fishing (150 m) was undertaken directly after eDNA sampling at three of the sampled sites with further three sites also compared (but not directly after sampling). The details (methods, results, and discussion) of these six sites comparing eDNA results with electric fishing are provided in Appendix S1. In summary, acknowledging differences between sampling distance of eDNA and electric fishing, a 32.6% increase in fish species richness was attained with eDNA sampling compared with electric fishing. However, among these six sites, two fish species not found with eDNA (six replicates) were found via electric fishing (sites fished directly after eDNA sampling), indicating that some rare, low biomass species may not be detected by eDNA metabarcoding. All other fish species (from the six sites) caught via electric fishing (95.3%) were detected with eDNA plus an additional 17 species across the six sites that were fished. This baseline data aligns with data collected by David et al. (2021) from five sites electric fished (single pass, 150 m) after three eDNA replicates, where a 29.3% increase in fish richness was observed from the eDNA sampling. Similarly, two species caught via electric fishing in low biomass were not detected in the eDNA metabarcoding results, though only three replicates were taken. More recently obtained data (J. Smith, unpublished data) from 44 stream sites (Waikato Region, stream order range 1–4) that were electric fished (single pass, 150 m) directly after eDNA sampling (six replicates) showed that 35.2% more species were detected with eDNA and that 5 species or 2% were not detected via eDNA metabarcoding. Again, as with other sampling events, these species detected via electric fishing and not by eDNA metabarcoding were rare with a low biomass.

3.6 | Effect of Elevation on Fish eDNA Replication

There was no significant difference (ANOVA, $df = 1$, $F = 1.42$, $n = 51$, $p = 0.239$) in proportional richness between small and larger waterbodies (stream order) for fish at 6 replicates, but there was a significant difference (ANOVA, $df = 1$, $F = 8.34$, $n = 51$, $p = 0.005$) between sites at low versus high elevation

(> 100 m), with sites at higher elevation starting at 70.4% species proportional richness at one replicate compared to lowland sites at 61.9% at one replicate (Table 3). For fish, the asymptote at these higher elevated sites is obtained faster (90.8% proportional fish richness after five replicates). In general, fish biodiversity throughout NZ decreases as elevation increases reflecting the elevation limited diadromy of NZ's native fish (McDowall 2000). Therefore, fish diversity is often much lower and homogenous for considerable distances at higher elevation sites. Sample replication could reduce again at even higher elevation, but the final choice of the number of replicates also needs to consider other factors such as sampling consistency, waterway size, and the likely amount of introduced species present. Similar results were seen by Bylemans et al. (2018) study where replication needed for fish varied along the altitudinal and biodiversity gradient in an Australian river. Sites above 100 m were limited to 13 in our study, but diadromous species often still occur at and above this elevation if the rise is gradual, and more work may be needed above elevations of around 200 m to assess optimal replication at these sites for fish in NZ.

3.7 | NEMS Taxa

The eDNA analysis originally identified 561 freshwater invertebrate taxa, but this list was reduced to 129 taxa (Table A4) when identified to the NEMS taxonomic level. The higher taxonomic richness for the NEMS invertebrates reflects the inherently larger species pool compared to fish, resulting in a higher species/taxa difference between replicates with an average of 10 taxa per replicate and a range of 3–17 (Table 3). Not surprisingly, there was a significant difference in the average taxonomic richness between sites when identified using either eDNA or kick nets to the NEMS standard or when using the original (mostly species) data from the eDNA results (ANOVA, $df=2$, $F=139.8$, $n=53$, $p<0.001$). Post hoc tests showed no differences in the taxonomic richness between taxon richness from kick nets or from the eDNA samples when identified to the NEMS level.

The identification of 561 freshwater invertebrate taxa from the eDNA highlights the ability of this technique to identify taxa to lower levels than those obtained by traditional morphological methods, which are often limited to relatively coarse taxonomic resolutions, for example, the NEMS protocols (NEMS 2022). Furthermore, Greenwood et al. (2015) developed an even more comprehensive list of 240 freshwater invertebrate taxa (identified to genus level where possible) that was gleaned from surveys at 1266 sites throughout NZ, which currently represents the best information to date on invertebrate distribution patterns throughout NZ. The eDNA survey of only 51 sites in this study identified more than double this number highlights the benefits of eDNA over traditional sampling, particularly in terms of undertaking proper biodiversity assessments. In particular, many invertebrate groups, that are currently identified only to coarse taxonomic levels such as oligochaetes, were found to have a very high diversity. The eDNA sampling thus recognized 58 different taxa within the class Oligochaeta, of which 26 were identified as species, while the other 32 taxa were identified as higher taxonomic levels (genera, family). Other diverse taxonomic groups where eDNA had a clear advantage over the

traditional NEMS level of identification included Cladocera (22 eDNA taxa), Copepoda (16 eDNA taxa), and Nematoda (18 eDNA taxa), where these were simply grouped into their single taxon under NEMS.

Despite the obvious increased ability of eDNA to identify taxa to lower taxonomic levels, we only assessed cumulative taxa richness plots at the same taxonomic resolution as used in the NEMS protocols. This allowed us to compare cumulative taxa richness plots collected by eDNA and traditional methods used in the NEMS protocols. Our results found that a single eDNA sample contained on average about 56% of the total taxonomic information that was found from 16 replicates and that collecting six replicates (the number we recommend for on-going eDNA surveys) could be expected to contain 86% of taxa at a site. These results are very different to those from data collected by traditional techniques. For example, Shaver and Suren (2011) examined the effect of increased sample replication on cumulative taxa richness curves from samples collected by using either a kick net or a Hess sampler in streams in Wellington, New Zealand. They found that cumulative richness tended to plateau between 8 and 15 replicate samples but that the first sample contained only about 25% of the total taxa richness found in the stream, based on 20 replicates. Collecting 6 replicate samples contained only about 60%–68% of total taxa richness from the kick or Hess samplers, respectively. A similar result was found by Suren and Lambert (2006), who found that a single kick sample contained about 39% of the total taxa richness in a small stream in New Zealand's west coast, while 6 replicates contained about 72% of total taxa richness.

These cumulative taxa curves from traditional morphological identification used in the NEMS protocols thus had a lower rate of taxa increase per replicate compared to that found in the eDNA samples. The higher initial number of taxa encountered using eDNA most likely reflects the much greater chance of DNA fragments that are lost from organisms upstream from the sampling site being collected with the syringes from the moving water (like fish). In contrast, animals collected with kick nets must be physically dislodged from the substrate in a stream and trapped in a downstream net. Given the often highly patchy nature of invertebrate distribution patterns (e.g., Winemiller, Flecker, and Hoeinghaus 2010), it is not surprising that taxa accumulation curves often start out with so few taxa in the first few samples.

National State of Environment (SoE) monitoring throughout NZ currently identifies invertebrates to the NEMS taxa list, which recognizes only 197 taxa. Even the more comprehensive list of 240 taxa developed by Greenwood et al. (2015) is far fewer than we found in this work. A key part of this biomonitoring work is the use of biotic indices such as the MCI, and its quantitative variant, the QMCI (Stark and Maxted 2007), or the Average Score per Metric (ASPM: Collier 2008). These are now recognized as compulsory attributes for councils to monitor under the new National Policy Statement for Freshwater Management (2020), a central government initiative to help improve water quality throughout NZ. The ASPM is the average of the % of sensitive Ephemeroptera, Plecoptera, and Trichoptera (EPT) abundance, EPT richness, and the MCI. This means that all three biotic indices used in the NPSFM (2020) rely to a very

large extent on the MCI. This index was originally developed to assess organic enrichment in streams and assigns individual taxa-specific tolerance scores based on their observed distributions above and below point source pollution sources from dairy farm milking sheds (Stark 1993). An obvious weakness of the MCI is that it has published tolerance scores only for very broad taxonomic groups of many freshwater groups. For example, the class Oligochaeta has a general tolerance score of 1, indicating a tolerance to organic enrichment. However, given the large diversity of Oligochaeta identified by eDNA (including 28 identified species), this is an obvious weakness, as many of these species are likely to have a more narrow tolerance to organic enrichment. The relatively rapid advancements in eDNA technology and creation of ever-expanding genetic reference databases like GenBank/BOLD suggests that eDNA may have an important role to play in future biomonitoring programmes throughout NZ. However, a major requirement in accepting this technology in biomonitoring programmes is to create a consistent methodology for sample collection. Based on the results of the cumulative taxonomic plots, a key methodological component is the acceptance of using six replicate samples to characterize the invertebrate communities, as this number of replicates was shown to detect the vast majority of taxa expected at a site. Finally, another obvious step in using eDNA for routine biomonitoring throughout NZ would be to develop an eDNA version of the MCI, using taxa identified to the lowest possible level. This would likely perform better the current version, where much information on the tolerances of different taxa is lost when groups are combined to class or higher. eDNA sampling also has additional advantages over traditional invertebrate collection methods (kick net/surber sampling) in which the quality of the data collected by traditional methods can often depend on the experience and skills of the staff involved in the collection and processing of the samples (Baird and Hajibabaei 2012; Uchida et al. 2020). eDNA monitoring can overcome some of these issues, by simply collecting a water sample. Moreover, the site is not disturbed, animals are not killed to identify them, sampling bias in different mesohabitats is reduced, and both lentic and lotic invertebrates are detected (Mächler, Osathanunkul, and Altermatt 2018). Finally, results would generally arrive much quicker than traditional sampling, and there is also likely to be fewer inconsistencies in identification compared with using morphological characteristics (Serrana et al. 2018).

3.8 | Replication Optimisation

This study as well as others (Mächler et al. 2014; Evans et al. 2017; Beentjes et al. 2019; Cantera et al. 2019; Macher et al. 2021, and David et al. 2021) showed a positive relationship between increased replication and/or water volume and increases in species richness. However, the extra work and cost involved in collecting and processing these samples need to be optimized, thus finding a balance between cost and comprehensive biodiversity assessment. Work by Macher et al. (2021) found fish species richness increased by 24.8% with 18 one liter water replicates from 4 sites along a 2 km stretch of the Mulde River in Dessau Germany. This is comparable to 25.4% for fish over six replicates or 36.0% for all 16 replicates in this study but with many more sites. Macher et al. (2021) also exhibited a similar

starting point for species richness after 1 replicate and recommended 8–10 replicates for fish fauna, with no substantial richness increase above these numbers. This high starting point for fish after one replicate was also seen by Cantera et al. (2019) for fish in tropical Guianese rivers and streams, where after one replicate (34 L per replicate) average species richness was 67% in rivers and 87% in streams. This compares to 64% for fish in this study, where only 1 L of water was filtered. After two replicates (or 68 L of water filtered), Cantera et al. (2019) found little additional species richness, leading to their conclusion that filtering 34–68 L of water was sufficient for inventorying the fish fauna in tropical streams.

Cantera et al. (2019) along with others (Pont et al. 2023) also suggested that more replicates may be needed for larger flowing waterbodies to describe the typically greater species diversity found in these larger systems. However, this does not appear to be the case in this study and in earlier work by David et al. (2021), with little difference being found between small streams (order 1–3) and larger streams and rivers (order 4–8). Another difficulty with using findings from other countries about defining the optimum volume of water to be filtered is that fish diversity is relatively low in NZ compared to many other countries, so less replication may in fact be required to adequately characterize the fish community, especially in larger water bodies. The same may also apply to the volume of water filtered, and the boosted method may show a different result in more diverse systems.

Differences between taxa groups based on diversity within a given system can also influence the number of replicates that might be needed to assess a community, especially if the focus is two or more taxa groups (Furlan et al. 2016). In this study, fish and the NEMS taxa are the main taxa of interest and there is a diversity difference, with fewer species for fish, 40 in total compared to 129 invertebrate taxa identified or < 10 fish species per site compared to 35+ invertebrate taxa. As a result, fish maintains slightly a higher level of proportional detection relative to NEMS taxa. However, this cannot entirely be attributed to diversity, and several other factors like biomass and shedding rates (habitat, physiology, behavior, and predation) will also account for differences between fish and invertebrate apparent richness (Allan et al. 2021; Deiner et al. 2017). Despite these differences, the level of replication needed to describe these two groups adequately in this study is similar, which is a useful outcome when designing sampling programs.

This study highlights the need for replication in the detection of rare or patchy species (Lopes et al. 2017; Cantera et al. 2019; Macher et al. 2021; Takahashi et al. 2023). However, replication for abundant/common species or near field species (species present just upstream of the sampling location) is less of an issue as they are easily detected compared to rarer species at the location sampled. Most sites in this study had some fish species present across all eight replicates with high sequence reads. These species at the electric fishing sites were considered common to abundant at this reach level and clearly near field when eDNA samples were taken. Detection of fish species between replicates in this study was stochastic, particularly for rare species and those with patchy distributions. It should also be noted

that this analysis is based on species/taxa being detected from the 16 replicates from each site, and it is possible that some species were entirely missed by eDNA sampling due to either being very rare or a species having low DNA shedding rates compared with other species (see Appendix S1 for fish). The best results for eDNA as with more traditional sampling methods will likely be during the warmer months (December–April) in NZ (Joy, David, and Lake 2013), when fish are more active, migrations are occurring, and water flows are generally lower and more stable (Furlan et al. 2016; Buxton, Groombridge, and Griffiths 2018). More replication could be required during colder months for effective community characterization (Furlan et al. 2016). Also, seasonality maybe more species specific in terms of detection with eDNA when spawning occurs (Hayami et al. 2020; Jo et al. 2021).

When trying to detect or locate very rare or newly invasive fish species more than six replicates may be required as several rare/patchily distributed species only occurred in the 7th or 8th replicate at some sites in this study. Metabarcoding can be used as a cost-effective means to gain a perspective on the various taxa present for broad biodiversity monitoring. It should be noted that targeted approaches such as qPCR and ddPCR can offer increased sensitivity for individual species when the risk tolerance for type II errors/false negatives is low (Wood et al. 2019). In these situations, such as biosecurity incursion responses and threatened species mapping, a combined general metabarcoding and targeted assay approach may be preferable.

After discussion with stake holders involved in this study, six replicates were recommended as the “A” standard for eDNA sampling for freshwater fish and NEMS taxa in NZ. Any fish species undetected with six replicates are likely to be either rare or located a considerable distance upstream of the sampling point (dependant on flow).

4 | Conclusions

The benefits of eDNA sampling by agencies throughout NZ to describe the overall species composition in waterways are obvious. However, the ultimate use of eDNA is likely to be site specific and depends on the questions being asked. Undertaking biodiversity assessments using eDNA will, in most cases, be easier with eDNA sampling compared to more labour-intensive traditional methods when the presence/absence data is required. A core thrust of this work was consequently to develop a standardized protocol for the collection of eDNA throughout NZ that regional councils and other agencies can use. Although we have also suggested that there may be instances where six replicates may be considered unnecessary (e.g., to document fish diversity in high elevation inland waterways), it is also considered important that we support a standardized monitoring program throughout the country. Collecting the full set of 6 replicates at all sites is likely to provide more valuable community information, for aquatic communities and also some terrestrial species, as well as allowing us to better describe overall stream health using novel approaches such as the Taxon-Independent Community Index (TICI), (Wilkinson et al. 2024) or future prevalence

indicators based on sequence reads that can be generated from 6-replicate data. Although there are now established biotic metrics to assess fish community health (Fish IBI: Joy, David, and Lake 2013) and invertebrate community health (MCI: Stark and Maxted 2007), these metrics (especially for macro-invertebrates) are unlikely to be applicable to the full suite of taxonomic data collected by eDNA. An obvious next step in enhancing the interpretative power of eDNA is to establish an eDNA Fish-IBI (Index of Biological Integrity) and an eDNA MCI (Macroinvertebrate Community Index) based on NEMS taxa for environmental reporting throughout NZ. Although eDNA has an obvious advantage in obtaining species composition data and, to a lesser extent, relative abundance data using the “read counts”, it does have its limitations. For example, traditional methods can measure fish and macroinvertebrate abundance and provide fish size structure. It is unlikely that eDNA will be able to provide these types of information. However, eDNA is a rapidly evolving field, and the current methods used in this study are likely to evolve into new and more efficient methods that may be able to solve some of these issues. Despite the undoubtful huge progress made to date with eDNA technology, there are several uncontrollable factors in the various stages of eDNA collection and analysis that need to be minimized to obtain the best results. This can only be achieved by adopting sound standard practices both in the field and the laboratory, which is why we recommend collecting 6 replicate syringe samples for ongoing eDNA surveys throughout NZ.

Author Contributions

Conception or design of the study: J.S., B.D., A.H., S.W.; The acquisition, analysis, or interpretation of the data: J.S., B.D., S.W.; Writing of the manuscript: J.S., B.D., A.H., S.W., N.L., A.S., A.G., D.F.

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Ethics Statement

This manuscript has not been submitted elsewhere. No ethics approval needed for this study as mainly eDNA sampling only. Ethics for the limited electric fishing sites are covered under Waikato Regional Councils Special Permit (705-3 Fisheries New Zealand, Ministry of Primary Industries) and Authority to Operate an Electric Fishing Device (Department of Conservation, Authorisation Number: 98118-RES).

Conflicts of Interest

Shaun Wilkinson, a co-author of this manuscript, is the Director of Wilderlab where samples for this project were analyzed.

Data Availability Statement

The raw data from this study has been uploaded to SRA with the following link: <https://www.ncbi.nlm.nih.gov/sra/PRJNA950216>. Standard Bioinformatic processing script is available with the following link: [Laulala333/High Replication paper Bioinformatics.R at main · Laula la333/Laulala333\(github.com\)](#).

Permission to Reproduce Material From Other Sources

All participants (authors and Councils who contributed to the data) have given their permission for this study to be published.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

TABLE A1 | Additional details for national high replicate eDNA sites ($n = 54$), sampled December 2020 to May 2021.

Map ID	Site ID	Job ID	Territorial authority	Site name	Lat	Lon	Collection date	Standard (mL)	Boosted (mL)	Substrate	Elevation (m)	Stream order	NZSEGMENT	NZREACH	FENZ LEVEL FOUR ALPHA	Dominant land use (5km)
1	RC031	601114	Northland Regional Council	Tapapa Stream	-35.19295451	173.4806506	2021-01-28	1000	2000	HB	80.4	3	1006293	1005772	C5.1a	Native vegetation
2	RC029	601109	Northland Regional Council	Waipoua River	-35.65249605	173.570394	2021-01-29	850-1000	1800-2000	HB	99.6	4	1018149	1016579	C5.1b	Native vegetation
3	RC027	601111	Northland Regional Council	Waiponamu stream	-35.74427694	174.2931542	2021-02-02	1000	2000	HB	58.5	3	1020553	1018851	C5.1a	High pastoral
4	RC028	601112	Northland Regional Council	Tangowahine Stream	-35.84428975	173.9359479	2021-02-02	1000	2000	HB	18.8	5	1022656	1020873	C4.2b	High pastoral
5	RC030	601113	Northland Regional Council	Ruakaka River	-35.86944231	174.4026121	2021-01-27	650-800	1320-1800	SB	17.6	4	1023322	1021503	A3.1b	High pastoral
6	RC035	601289	Auckland Council	Avondale Stream	-36.923647	174.69076	2021-03-17	300-400	310-800	SB	20.0	3	2039302	2006297	A1.1a	Urban
7	RC036	601287	Auckland Council	Piha Stream	-36.95342	174.475644	2021-03-10	500-1000	1400-2000	HB	19.6	3	2039826	2006509	C5.1a	Native vegetation
8	RC037	601288	Auckland Council	Ngakoroa Stream	-37.19548	174.973634	2021-03-09	550-1000	1450-2000	SB	159.1	3	2045595	2009507	A4.2a	High pastoral
9	RC006	601050	Waikato Regional Council	Komakorau Stream	-37.699054	175.329883	2021-01-20	160-200	260-475	SB	30.1	4	3059477	3015932	B1.2a	High pastoral
10	RC051	601611	Bay of Plenty Regional Council	Whanarua Stream	-37.68458853	177.7887414	2021-05-11	1000	2000	HB	24.0	2	4062208	4001746	C5.1a	Native vegetation
11	RC005	601049	Waikato Regional Council	Waikato River @ Hamilton	-37.793849	175.290637	2021-01-25	1000	2000	HB/SB	20.2	7	3063520	3017901	A1.2a	Urban
12	RC039	601551	Bay of Plenty Regional Council	Mimiha Stream	-37.8672189	176.6946157	2021-04-23	1000	2000	SB	16.5	3	4070381	4004561	C6.1a	High Pastoral
13 ^a	RC038	601552	Bay of Plenty Regional Council	Karaponga Stream ^a	-38.00962068	176.7079927	2021-04-30	1000	2000	SB	39.0	3	4079858	4008127	C4.2c	High pastoral
14	RC050	601609	Bay of Plenty Regional Council	Ohutu Stream	-38.39083617	176.8317664	2021-05-10	1000	2000	HB	241.9	3	4112447	4021130	C12.1b	Native Vegetation
15	RC052	601239	Waikato Regional Council	Manganui Trib @ Small 3	-38.51715248	174.7105166	2020-12-14	1000	2000	HB	180.0	3	3118742	3037993	C7.1c	Native vegetation
16	RC053	601240	Waikato Regional Council	Manganui River @ Big 3	-38.51715248	174.7105166	2020-12-14	1000	2000	HB	180.0	3	3118742	3037993	C7.1c	Native vegetation
17	RC054	601241	Waikato Regional Council	Manganui River @ 410_8	-38.59365209	174.6835227	2020-12-15	1000	2000	HB	60.4	4	3125700	3040629	C6.2a	native vegetation
18	RC010	601330	Hawke's Bay Regional Council	Nuhaka River	-39.005279	177.764983	2021-03-31	1000	2000	HB	20.4	4	8167155	8012981	C4.2d	High pastoral

(Continues)

TABLE A1 | (Continued)

Map ID	Site ID	Job ID	Territorial authority	Site name	Lat	Lon	Collection date	Standard (mL)	Boosted (mL)	Sub-strate	Elevation (m)	Stream order	NZSEGMENT	NZREACH	FENZ LEVEL FOUR ALPHA	Dominant land use (5km)
19	RC007	601096	Taranaki Regional Council	Maketawa Stream	-39.2322104	174.1843902	2021-02-01	1000	2000	HB	375.3	2	6178774	6005874	C2.1a	Native vegetation
20	RC011	601286	Hawkes Bay Regional Council	Waingongoro Stream	-39.382179	176.3325	2021-03-19	1000	2000	HB	497.9	3	8192205	8021421	C6.4a	High pastoral
21	RC008	601403	Taranaki Regional Council	Kaupokonui Stream	-39.54029426	174.0826127	2021-04-15	1000	2000	HB	59.8	5	6198171	6011674	C2.1a	High pastoral
22	RC009	601404	Taranaki Regional Council	Tangahoe River	-39.62909427	174.3490832	2021-04-15	510-700	1070-1210	HB	21.2	5	6206136	6014041	A4.2a	High pastoral
23	RC014	601158	Hawkes Bay Regional Council	Ngauroro River @ Fernhill	-39.58803	176.769977	2021-03-21	1000	2000	HB	20.0	6	8207185	8024814	C4.2d	High pastoral
24	RC012	601277	Hawkes Bay Regional Council	Kuripapango Stream	-39.807778	176.98567	2021-03-21	1000	2000	HB	16.5	1	8220246	8028428	A1.1b	Native vegetation
25	RC013	601323	Hawkes Bay Regional Council	Papanui Stream	-39.855926	176.715002	2021-03-26	1000	2000	SB	80.0	5	8221860	8029023	A4.1c	High pastoral
26	RC025	601182	Horizons Regional Council	Tamaki River @ Reserve	-40.1200834	176.031067	2021-02-24	1000	2000	HB	401.0	3	7231345	7034557	C7.1d	Native vegetation
27	RC026	601245	Horizons Regional Council	Turitea Stream @ Massey Farm Bridge	-40.385336	175.607949	2021-03-17	1000	2000	HB	39.0	3	7239293	7040071	C6.4a	High pastoral
28	RC024	601216	Horizons Regional Council	Waikawa Stream @ North Manakau Road	-40.706966	175.23451	2021-03-10	1000	2000	HB	61.2	4	7248627	7048450	C7.1c	High pastoral
29 ^a	RC048	601400	Greater Wellington Regional Council	Whareroa Stream ^a	-40.955839	174.970188	2021-04-16	500	1000	SB	13.6	4	9255944	9005999	A4.2a	High pastoral
30	RC046	601337	Greater Wellington Regional Council	Waipoua River	-40.953588	175.6729873	2021-04-07	1000	2000	HB	104.0	5	9256259	9006301	C6.4a	High pastoral
31	RC049	601378	Greater Wellington Regional Council	Wainui Stream	-40.9739501	174.9616078	2021-04-13	1000	2000	HB	12.1	3	9256465	9006480	A4.2a	Native vegetation
32	RC023	601162	Nelson City Council	Whangamoa River @ Kokorua	-41.129	173.541	2021-02-23	1000	2000	HB	19.9	4	10009423	10007714	C6.4b	Forestry
33	RC021	601252	Nelson City Council	Wakapuaka River @ Hira	-41.214	173.399	2021-03-18	1000	2000	HB	49.6	4	10012362	10009633	C6.4b	Forestry
34	RC047	601520	Greater Wellington Regional Council	Kaiwharawhara Stream	-41.259917	174.788556	2021-05-07	1000	2000	HB	19.7	4	9263955	9013529	C5.2a	Native vegetation
35	RC020	601102	Nelson City Council	Maitai River @ Groom	-41.287	173.328	2021-02-04	1000	2000	HB	47.1	4	10014643	10011156	C6.4b	Forestry
36	RC018	601082	Nelson City Council	Poorman Valley Stream @ Seaview	-41.303	173.226	2021-02-03	1000	2000	HB	4.8	2	10016187	10011985	C5.2b	High pastoral

(Continues)

TABLE A1 | (Continued)

Map ID	Site ID	Job ID	Territorial authority	Site name	Lat	Lon	Collection date	Standard (mL)	Boosted (mL)	Substrate	Elevation (m)	Stream order	NZSEGMENT	NZREACH	FENZ LEVEL FOUR ALPHA	Dominant land use (5 km)
37	RC019	601098	Nelson City Council	Poorman Valley Stream @ Barnicoat	-41.317	173.25	2021-02-04	1000	2000	HB	74.6	2	10016393	10012419	C5.2a	Native vegetation
38	RC022	601354	Nelson City Council	Roding River @ Caretaker	-41.357	173.26	2021-04-09	1000	2000	HB	202.7	4	10018126	10013741	C7.1c	Native vegetation
39	RC032	601164	Marlborough District Council	Are Are Creek	-41.46738	173.8264	2021-02-24	1000	2000	HB	33.8	4	11022708	11017507	C6.4b	Forestry
40	RC042	601479	Environment Canterbury	Blue Duck Creek	-42.27923926	173.7669203	2021-03-30	1000	2000	HB	14.8	2	13064994	13006067	C1.1b	Native vegetation
41	RC015	601203	West Coast Regional Council	Redjacks Creek	-42.437996	171.478171	2021-03-01	1000	2000	HB	81.2	3	12073197	12027614	C2.1b	Native Vegetation
42	RC040	601478	Environment Canterbury	Landslip Creek	-42.69208608	172.0452815	2021-03-31	1000	2000	HB	592.0	3	13084962	13017470	G3.1a	Native vegetation
43	RC016	601204	West Coast Regional Council	Okutua Stream	-43.200577	170.266371	2021-02-21	1000	2000	HB	20.2	2	12111395	12041149	C2.2a	Native vegetation
44 ^a	RC041	601477	Environment Canterbury	Fork Stream, Talus Spring ^a	-43.85172455	170.3522441	2021-04-22	1000	2000	HB	1052.1	3	13146902	13504544	H6.4a	High pastoral
45	RC033	601131	Environment Canterbury	Lower Rakaia	-43.89980816	172.2071035	2021-02-02	1000	2000	HB	2.4	7	13300010	13058524	A4.4a	High pastoral
46	RC017	601207	West Coast Regional Council	Haast River	-43.937771	169.295797	2021-02-19	1000	2000	HB	73.4	6	12152655	12052210	J8.3a	Native vegetation
47	RC034	601132	Environment Canterbury	Lower Ashburton @ Hakatere	-44.05245243	171.8045024	2021-02-05	1000	2000	HB	7.7	6	13155682	13061477	A4.2a	High pastoral
48	RC001	601130	Environment Canterbury	Lower Rangitata River	-44.18753881	171.5084259	2021-02-09	1000	2000	HB	2.2	6	13163160	13063572	A4.4a	High pastoral
49	RC044	601504	Otago Regional Council	Dunstan Creek	-44.93954022	169.7651639	2021-05-03	1000	2000	HB	430.2	5	14214374	14022310	D4.1a	High pastoral
50	RC043	601503	Otago Regional Council	Kauru River	-45.10909124	170.7417559	2021-04-22	1000	2000	HB	100.0	5	14224934	14029391	C8.4a	High pastoral
51	RC004	601316	Environment Southland	Oreti River at McKellar's Flat	-45.32618168	168.208478	2021-03-24	1000	2000	HB	599.4	5	15245827	15016746	G1.3a	Native vegetation
52	RC045	601505	Otago Regional Council	Clutha River @ Clydevale	-46.10141607	169.5326871	2021-05-04	1000	2000	HB/SB	20.1	8	14299081	14067409	A5.2a	High pastoral
53	RC003	601315	Environment Southland	Oreti River at Branxholme	-46.29545184	168.2841205	2021-03-23	1000	2000	HB	10.6	7	15312337	15057362	G1.4c	High pastoral
54	RC002	601317	Environment Southland	Waituna Creek at White Pine Road	-46.52024252	168.5427042	2021-03-23	200-250	377-530	HB	10.0	1	15318961	15061719	B2.1a	High pastoral

Abbreviations: Substrate HB, hard bottom; SB, soft bottom.
^aSites excluded from the data analysis.

TABLE A2 | Group reads per method.

Group	Standard reads	Boosted reads	Combined reads	% of total reads
Other	6,763,806	6,467,960	13,231,766	33.5658
Plants	3,084,086	2,996,699	6,080,785	15.4255
Fish	2,358,506	2,148,401	4,506,907	11.4329
Worms	1,877,277	1,862,996	3,740,273	9.4882
Ciliates	1,284,185	1,168,214	2,452,399	6.2211
Insects	861,423	883,224	1,744,647	4.4258
Mollusks	737,269	742,786	1,480,055	3.7545
Birds	662,483	602,777	1,265,260	3.2097
Mammals	384,444	418,729	803,173	2.0375
Heterokont algae	393,767	392,013	785,780	1.9933
Cnidarians	390,312	388,388	778,700	1.9754
Diatoms	313,791	303,202	616,993	1.5652
Green algae	224,978	183,806	408,784	1.0370
Crustaceans	176,491	182,243	358,734	0.9100
Red algae	112,961	105,783	218,744	0.5549
Fungi	115,854	98,939	214,793	0.5449
Rotifers	93,648	107,914	201,562	0.5113
Bacteria	77,645	54,406	132,051	0.3350
Cryptomonads	45,417	50,312	95,729	0.2428
Bryozoans	47,308	48,246	95,554	0.2424
Flatworms	27,573	22,610	50,183	0.1273
Springtails	24,891	24,703	49,594	0.1258
Mosses	21,666	19,669	41,335	0.1049
Amoebae	14,681	12,962	27,643	0.0701
Mites and ticks	10,027	11,066	21,093	0.0535
Liverworts	5612	4720	10,332	0.0262
Centipedes	1555	1852	3407	0.0086
Spiders	1081	888	1969	0.0050
Amphibians	726	809	1535	0.0039
Hornworts	90	350	440	0.0011
Lizards and snakes	67	77	144	0.0004

TABLE A3 | Freshwater fish species found in the eDNA samples ranked in order of highest combined reads with conservation status (Dunn et al. 2018).

Species name	Common name	Reads (standard)	Reads (boosted)	Reads (combined)	Site occurrence	Average reads per site	Status
<i>Anguilla dieffenbachii</i>	Longfin eel	398,980	450,437	849,417	51	16,655	At Risk
<i>Gobiomorphus breviceps</i>	Upland bully	152,626	172,872	325,498	17	19,147	Not Threatened
<i>Gobiomorphus huttoni</i>	Redfin bully	128,336	140,120	268,456	33	8135	Not Threatened
<i>Anguilla australis</i>	Shortfin eel	124,437	120,925	245,362	43	5706	Not Threatened
<i>Salmo trutta</i> ^a	Brown trout ^a	86,070	94,462	180,532	32	5642	Sports fish
<i>Retropinna retropinna</i>	Common smelt	80,397	94,290	174,687	21	8318	Not Threatened
<i>Galaxias maculatus</i>	Inanga	81,296	81,318	162,614	28	5808	At Risk
<i>Cheimarrichthys fosteri</i>	Torrentfish	78,137	69,605	147,742	29	5095	At Risk
<i>Galaxias brevipinnis</i>	Koaro	42,155	61,306	103,461	20	5173	At Risk
<i>Oncorhynchus mykiss</i> ^a	Rainbow trout ^a	45,395	47,279	92,674	13	7129	Sports fish
<i>Gobiomorphus hubbsi</i>	Bluegill bully	48,055	44,609	92,664	17	5451	At Risk
<i>Galaxias divergens</i>	Dwarf galaxias	29,149	30,639	59,788	4	14,947	At Risk
<i>Galaxias fasciatus</i>	Banded kokopu	20,680	26,487	47,167	13	3628	Not Threatened
<i>Gobiomorphus basalis</i>	Crans bully	9002	20,153	29,155	3	9718	Not Threatened
<i>Gambusia affinis</i> ^a	Mosquitofish ^a	16,809	12,102	28,911	6	4819	Pest
<i>Cyprinus rubrofasciatus</i> ^a	Koi carp ^a	10,181	17,762	27,943	2	13,972	Pest
<i>Carassius auratus</i> ^a	Goldfish ^a	13,282	11,817	25,099	7	3586	Pest
<i>Gobiomorphus cotidianus</i>	Common bully	11,139	6934	18,073	4	4518	Not Threatened
<i>Mugil cephalus</i>	Gray mullet	4368	8883	13,251	5	2650	Not Threatened
<i>Arripis trutta</i>	Kahawai	7189	5149	12,338	4	3085	Not Threatened
<i>Galaxias argenteus</i>	Giant kokopu	5310	4052	9362	6	1560	At Risk
<i>Galaxias vulgaris</i>	Common river galaxias/Canterbury galaxias	2691	3725	6416	2	3208	At Risk
<i>Geotria australis</i>	Pouched lamprey	2852	3317	6169	19	325	Threatened
<i>Perca fluviatilis</i> ^a	European perch ^a	367	5552	5919	5	1184	Pest/Sports fish
<i>Gobiomorphus gobioides</i>	Giant bully	1627	1879	3506	10	351	At Risk
<i>Galaxias southern</i>	Southern galaxias	926	1277	2203	1	2203	Threatened
<i>Rhombosolea retiarda</i>	Black Flounder	1249	759	2008	5	402	Not Threatened

(Continues)

TABLE A3 | (Continued)

Species name	Common name	Reads (standard)	Reads (boosted)	Reads (combined)	Site occurrence	Average reads per site	Status
<i>Scardinius erythrophthalmus</i> ^a	Rudd ^a	417	1445	1862	3	621	Pest/Sports fish
<i>Galaxias postvectis</i>	Shortjaw kokopu	856	767	1623	6	271	Threatened
<i>Galaxias cobitinis</i>	Lowland longjaw galaxias	1052	542	1594	1	1594	Threatened
<i>Stokellia anisodon</i>	Stokell's smelt	668	423	1091	2	546	At Risk
<i>Oncorhynchus tshawytscha</i> ^a	Chinook salmon/King salmon ^a	593	200	793	6	132	Sports fish
<i>Galaxias gollumoides</i>	Gollum galaxid	298	400	698	2	349	Threatened
<i>Aldrichetta forsteri</i>	Yelloweye mullet	249	286	535	4	134	Not Threatened
<i>Anguilla reinhardtii</i> ^b	Speckled longfin eel ^b	47	407	454	5	91	Non-Resident native
<i>Galaxias paucispindylus</i>	Alpine galaxias	228	187	415	2	208	Threatened
<i>Galaxias anomalus</i>	Roundhead galaxias	189	104	293	1	293	Threatened
<i>Forsterygion nigripenne</i>	Estuarine triplefin	154	0	154	1	154	Not Threatened
<i>Ameiurus nebulosus</i> ^a	Brown bullhead catfish ^a	74	66	140	1	140	Pest
<i>Galaxias</i> sp. D/ <i>Pomahaka Galaxias</i>	Clutha flathead/Pomahaka galaxias	0	24	24	1	24	Threatened

^aIntroduced.

^bNon-resident native.

TABLE A4 | Freshwater invertebrate species/taxa (Stark taxa) found in the eDNA samples ranked in order of highest combined reads.

Common name	Reads (standard)	Reads (boosted)	Reads (combined)	Site occurrence	Average reads per site
Oligochaeta	1,032,099	1,075,289	2,107,388	51	41,321
Hydra	381,556	374,549	756,105	51	14,826
Potamopyrgus	339,072	359,423	698,495	50	13,970
Cladocera	63,726	45,787	109,513	42	2607
Deleatidium	37,529	34,065	71,594	33	2170
Copepoda	31,686	31,637	63,323	47	1347
Nematoda	29,062	27,561	56,623	51	1110
Hydropsyche	31,090	22,315	53,405	27	1978
Nemertea	28,311	23,632	51,943	41	1267
Collembola	24,676	24,323	48,999	45	1089
Pycnocentroides	27,813	19,334	47,147	41	1150
Coloburiscus	14,822	17,565	32,387	33	981
Olinga	8577	13,814	22,391	26	861
Austrosimulium	9926	7477	17,403	30	580
Physella	8224	8633	16,857	26	648
Nesameletus	7081	6964	14,045	18	780
Acarina	5456	7158	12,614	34	371
Hydrobiosis	5990	6216	12,206	35	349
Chironomus	4319	6816	11,135	38	293
Atalophlebioides	5011	5238	10,249	11	932
Austroperla	4274	5321	9595	18	533
Hudsonema	4055	3423	7478	29	258
Bryozoa	3907	2738	6645	16	415
Platyhelminthes	3606	2887	6493	41	158
Chironomidae	1393	3355	4748	43	110
<i>Paratya curvirostris</i>	1589	1798	3387	14	242
<i>Austroclima sepia</i>	1479	1266	2745	9	305
Nesameletidae	967	1416	2383	6	397
Dixidae	1607	656	2263	16	141
<i>Aphrophila</i> sp. EJD-2015	1213	980	2193	7	313
Triplectides	969	1044	2013	13	155
Psilochorema	997	927	1924	24	80
Polypsectropus	991	911	1902	29	66
Zephlebia	1044	790	1834	7	262
Ameletopsis	1050	761	1811	11	165
Ichthybotus	589	1200	1789	11	163
Sphaeriidae	656	943	1599	15	107
Syrphidae	711	749	1460	20	73
<i>Sororipyrgus kutukutu</i>	626	779	1405	2	703

(Continues)

TABLE A4 | (Continued)

Common name	Reads (standard)	Reads (boosted)	Reads (combined)	Site occurrence	Average reads per site
Leptophlebiidae	632	668	1300	12	108
Gyraulus	434	823	1257	11	114
Culicidae	585	646	1231	7	176
Latia	686	541	1227	8	153
Paracalliope	77	820	897	3	299
Rallidens	576	309	885	8	111
Tardigrada	444	430	874	12	73
Spaniocercoides	487	349	836	1	836
Hirudinea	739	50	789	5	158
Zelandobius	375	399	774	11	70
Paranephrops	343	339	682	8	85
Elmidae	476	109	585	9	65
<i>Hygraula nitens</i>	279	255	534	6	89
Oxyethira	441	83	524	12	44
<i>Hydraenidae</i> sp. EJD-2015	145	372	517	12	43
Paroxyethira	275	220	495	4	124
Neurochorema	288	203	491	10	49
Melanopsis	272	161	433	4	108
Costachorema	171	225	396	14	28
Amphipoda	170	182	352	6	59
Ostracoda	169	170	339	12	28
Echyridella	104	210	314	3	105
Confluens	111	160	271	4	68
<i>Antipodochlora braueri</i>	71	190	261	6	44
Hydrobiosidae	117	124	241	6	40
Cryptobiosella	74	139	213	2	107
Diplectrona	59	151	210	2	105
Rhabdocoela	104	101	205	5	41
Beraeoptera	79	88	167	5	33
Zelandoperla	101	63	164	9	18
<i>Obtusopyrgus alpinus</i>	0	150	150	1	150
Archichauliodes	133	15	148	6	25
Hydroptilidae	107	34	141	3	47
Oeconesidae	20	121	141	4	35
Hydrobiosella	74	66	140	4	35
Stenoperla	54	83	137	7	20
Pycnocentria	96	40	136	8	17
<i>Orthopsyche fimbriata</i>	93	38	131	4	33
Isopoda	55	74	129	7	18

(Continues)

TABLE A4 | (Continued)

Common name	Reads (standard)	Reads (boosted)	Reads (combined)	Site occurrence	Average reads per site
Oeconesus	87	37	124	6	21
Helicopsyche	46	70	116	4	29
Triplectidina	0	114	114	2	57
Mauiulus	78	30	108	4	27
<i>Neocurupira</i> sp. EJD-2015	52	54	106	1	106
Spaniocerca	79	4	83	4	21
Orthoclaadiinae	32	50	82	5	16
Collembola	34	44	78	9	9
Rakiura vernale	33	40	73	1	73
<i>Glyptophysa</i> sp. NZla2	39	31	70	3	23
Lymnaeidae	0	68	68	3	23
Hydrophilidae	57	8	65	7	9
Staphylinidae	43	17	60	5	12
Xanthocnemis	42	13	55	5	11
<i>Stictocladus</i> sp. EJD-2015	25	28	53	2	27
Oniscigaster	49	3	52	3	17
Tabanidae	29	23	52	4	13
Philorheithrus	42	3	45	4	11
Oecetis	23	16	39	3	13
<i>Zelandoptila moselyi</i>	22	16	38	2	19
Notonemouridae	5	32	37	4	9
Plectrocnemia	13	23	36	3	12
Sigara	15	20	35	4	9
Tanytarsus	10	25	35	3	12
<i>Hydrochorema crassicaudatum</i>	16	18	34	2	17
Limonia	34	0	34	1	34
Austrolestes	22	8	30	3	10
Procordulia	9	18	27	1	27
Hemicordulia	0	23	23	3	8
Acroperla	0	22	22	1	22
<i>Nesoperla fulvescens</i>	19	0	19	1	19
Polypedilum	18	0	18	1	18
Zelolessica	16	0	16	1	16
Ephydriidae	0	15	15	2	8
Ceratopogonidae	6	6	12	3	4
<i>Rhantus suturalis</i>	8	3	11	1	11
Psychodidae	4	5	9	2	5
Corynoneura	0	6	6	1	6

(Continues)

TABLE A4 | (Continued)

Common name	Reads (standard)	Reads (boosted)	Reads (combined)	Site occurrence	Average reads per site
<i>Ischnura aurora</i>	0	6	6	1	6
<i>Alloecentrella magnicornis</i>	5	0	5	1	5
Ferrissia	5	0	5	1	5
Gundlachia	5	0	5	1	5
<i>Kempynus latiusculus</i>	0	5	5	1	5
Tiphobiosis	0	5	5	1	5
Megaleptoperla	4	0	4	1	4
Nematomorpha	0	4	4	1	4
Polypedilum	0	4	4	1	4
<i>Taraperla ancilis</i>	0	4	4	1	4
Aoteapsyche	0	3	3	1	3
Austronella	0	3	3	1	3
Stratiomyidae	0	3	3	1	3